Descriptive Statistics

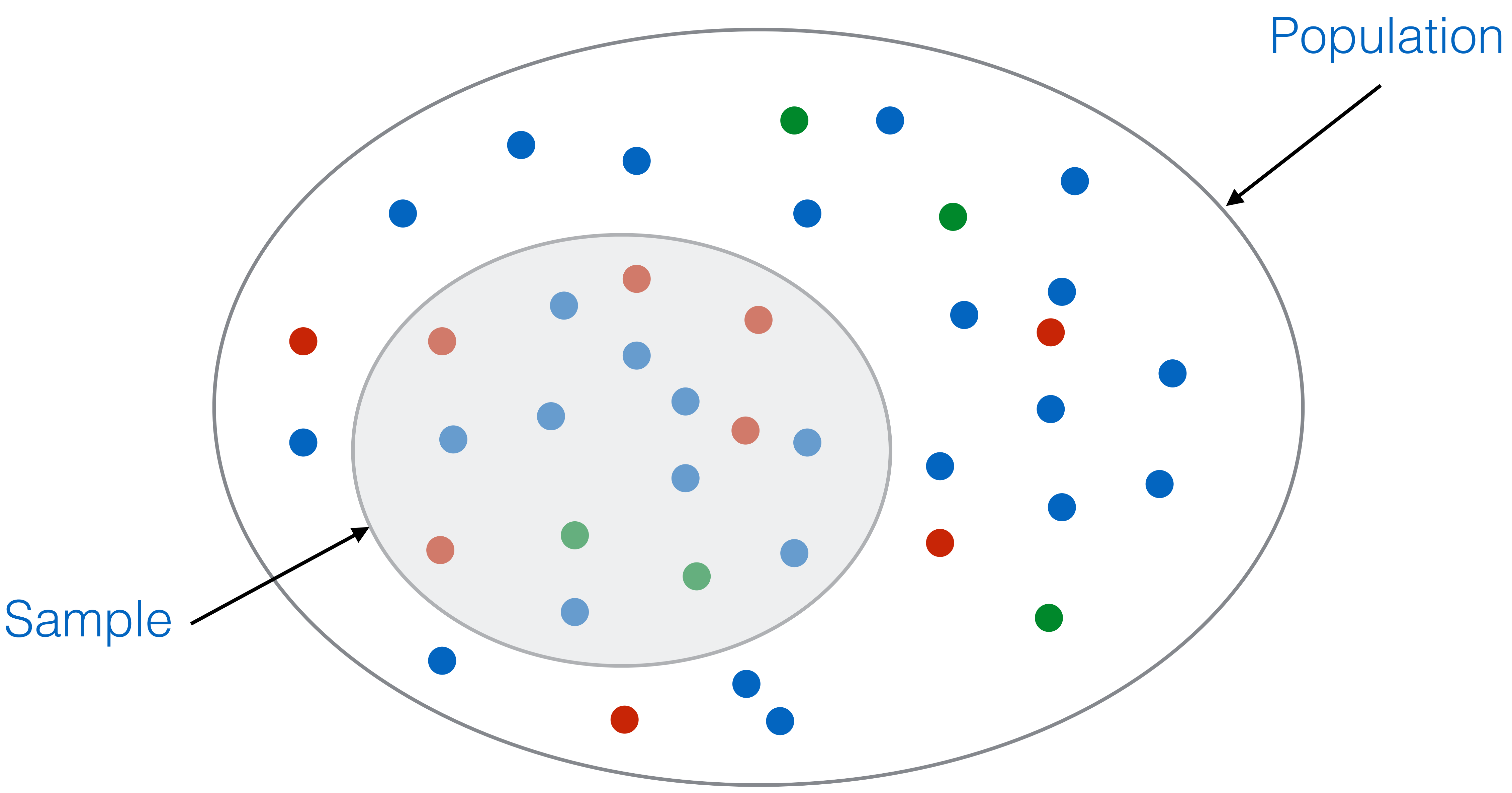
Lecture 8

Topics in this lecture

- Sampling
- Summary statistics
- Graphical summaries

Basic Idea

- Make inferences about a population by studying a relatively small sample chosen from it



Terminology

- *Population* — the entire collection of objects or outcomes about which information is sought
- *Sample* — a subset of a population, containing the objects or outcomes that are actually observed
- *Simple Random Sampling* — select a sample by a method in which each item is equally likely to be chosen (e.g., as in a lottery draw)

Types of Data

- Quantitative (numerical)
  - temperature, weight, number of daily visits on a website
- Qualitative (categorical)
  - gender (female, male)
  - shirt size (S, M, L, XL)

Sample Mean

- Also called the "arithmetic mean" or the "average"

- Let X_1, X_2, ..., X_n be a sample. The sample mean is

$$\boxed{\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i}$$

Sample Variance

- Let X_1, X_2, …, X_n be a sample. The sample variance is

$$\boxed{s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2}$$

- An equivalent formula, which can be easier to compute is

$$\boxed{s^2 = \frac{1}{n-1}\left(\sum_{i=1}^{n} X_i^2 - n\overline{X}^2\right)}$$

Sample Standard Deviation

- Let X_1, X_2, …, X_n be a sample. The sample standard deviation is

$$s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2}$$

- An equivalent formula is

$$s = \sqrt{\frac{1}{n-1}\left(\sum_{i=1}^{n}X_i^2 - n\overline{X}^2\right)}$$

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Example 1

Find sample mean, sample variance, and sample standard deviation of students' weight

Student Id	Weight (kg)
1	45
2	49
3	55
4	41
5	62

$$\overline{X} = \frac{1}{5}(45 + 49 + 55 + 41 + 62) = 50.4$$

$$s^2 = \frac{1}{n-1}\left(\sum_{i=1}^{n} X_i^2 - n\overline{X}^2\right)$$

$$\sum X_i^2 = (45^2 + 49^2 + 55^2 + 41^2 + 62^2)$$
$$= 12976$$

$$s^2 = \frac{1}{4}\left(12976 - (5 \times 50.4^2)\right) = 68.8$$

$$s = \sqrt{68.8} = 8.2946$$

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Modification of Data

- Let X_1, X_2, …, X_n be a sample, and $Y_i = aX_i + b$, where a and b are constants, then

$$\overline{Y} = a\overline{X} + b$$

$$s_Y^2 = a^2 s_X^2$$

$$s_Y = |a| s_X$$

Sample Median

- To find the median of a sample, order the values from smallest to largest. The sample median is the middle value

- If n numbers are ordered from smallest to largest

    - If n is odd, the sample median is the number in position $(n+1)/2$

    - If n is even, the sample median is the average of the numbers in positions $n/2$ and $n/2 + 1$

- The median is often used as a measure of center for samples that contain outliers

Example 2

Total scores made by Michael Jordan is shown in the following table, find the median

Year	Scores
1984	2313
1985	408
1986	3014
1987	2868
1988	2633
1989	2753
1990	2580
1991	2404
1992	2541
1994	457
1995	2491
1996	2431
1997	2357
2001	2375
2002	1640

$n = 15$

Mode and Range

- Mode — most frequently occurring value in the sample. If several values occur with equal frequency, each one is a mode
- Range — the difference between the largest and smallest values in a sample. It is a measure of spread, but it is rarely used because it depends only on the two extreme values and provides no information about the rest of the sample.

Quartiles

- Quartiles divide the sample as nearly as possible into quarters
- Simple way to compute quartiles of a sample of size *n*
  - Order the values from smallest to largest
  - To find the 1st quartile, compute the value 0.25(n+1). If this is an integer then the sample value in that position is the first quartile. If not, then take the average of the sample values on either side of this value.
  - The 2nd quartile is computed in the same way except that the value 0.50(n+1) is used.
  - The 3rd quartile is computed in the same way except that the value 0.75(n+1) is used.

Example 3

Total scores made by Michael Jordan is shown in the following table, find the 1st, 2nd, and 3rd quartiles

Year	Scores
1984	2313
1985	408
1986	3014
1987	2868
1988	2633
1989	2753
1990	2580
1991	2404
1992	2541
1994	457
1995	2491
1996	2431
1997	2357
2001	2375
2002	1640

$n = 15$

Percentiles

- The p^{th} percentile of a sample, for the number p between 0 and 100, divides the sample so that as nearly as possible p% of the sample values are less than the p^{th} percentile, and (100 - p)% are greater

- Simple way to compute percentiles of a sample of size *n*

  - Order the sample values from smallest to largest, and then computer the quantity $(p/100) \times (n+1)$

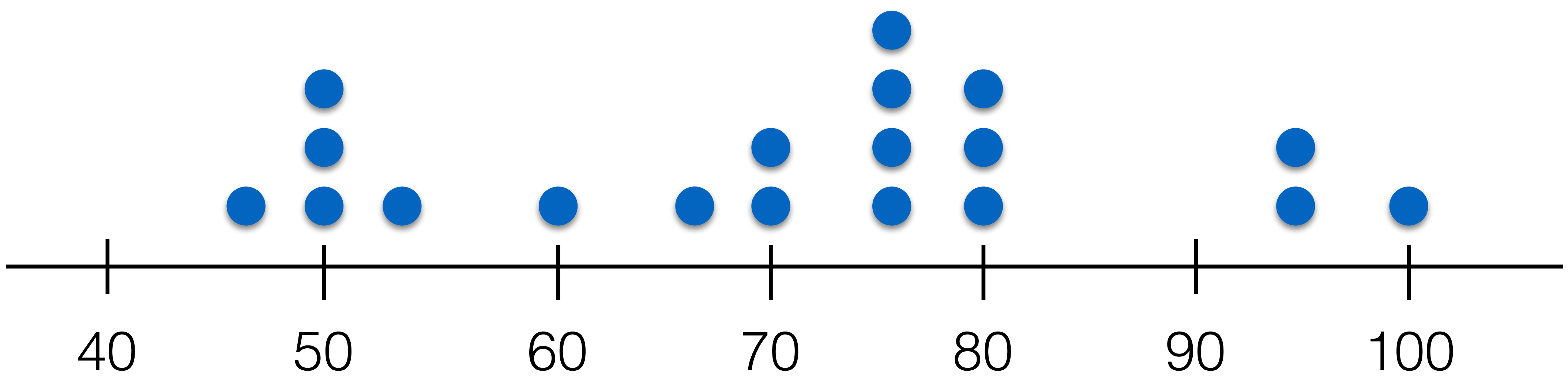
  - If this value is an integer, the sample value in this position is the p^{th} percentile. Otherwise average the two sample values on either side.

Graphical Summaries

- Dotplots
- Histograms
- Boxplots

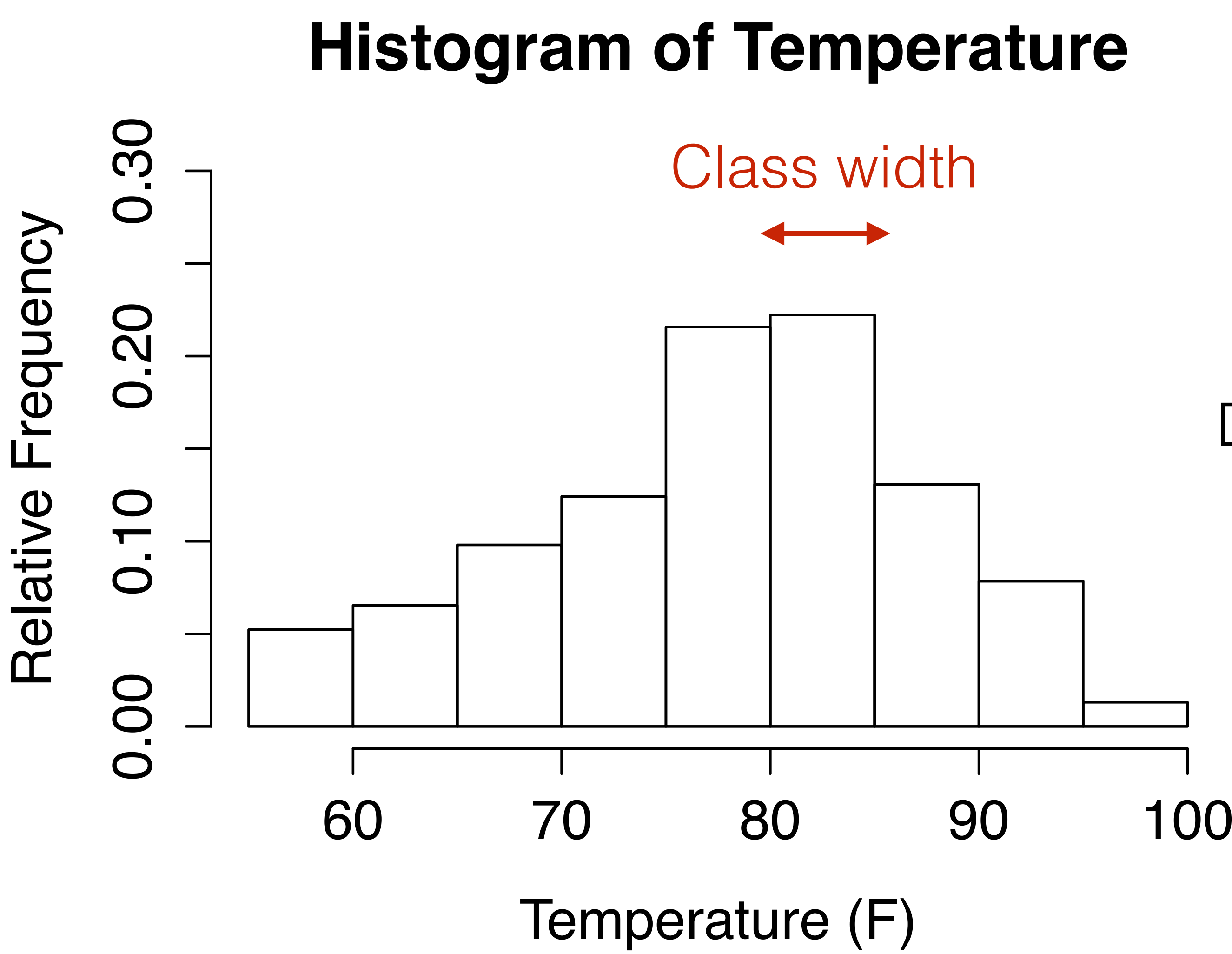
Dotplots

- Can be used to give a rough impression of the shape of a sample



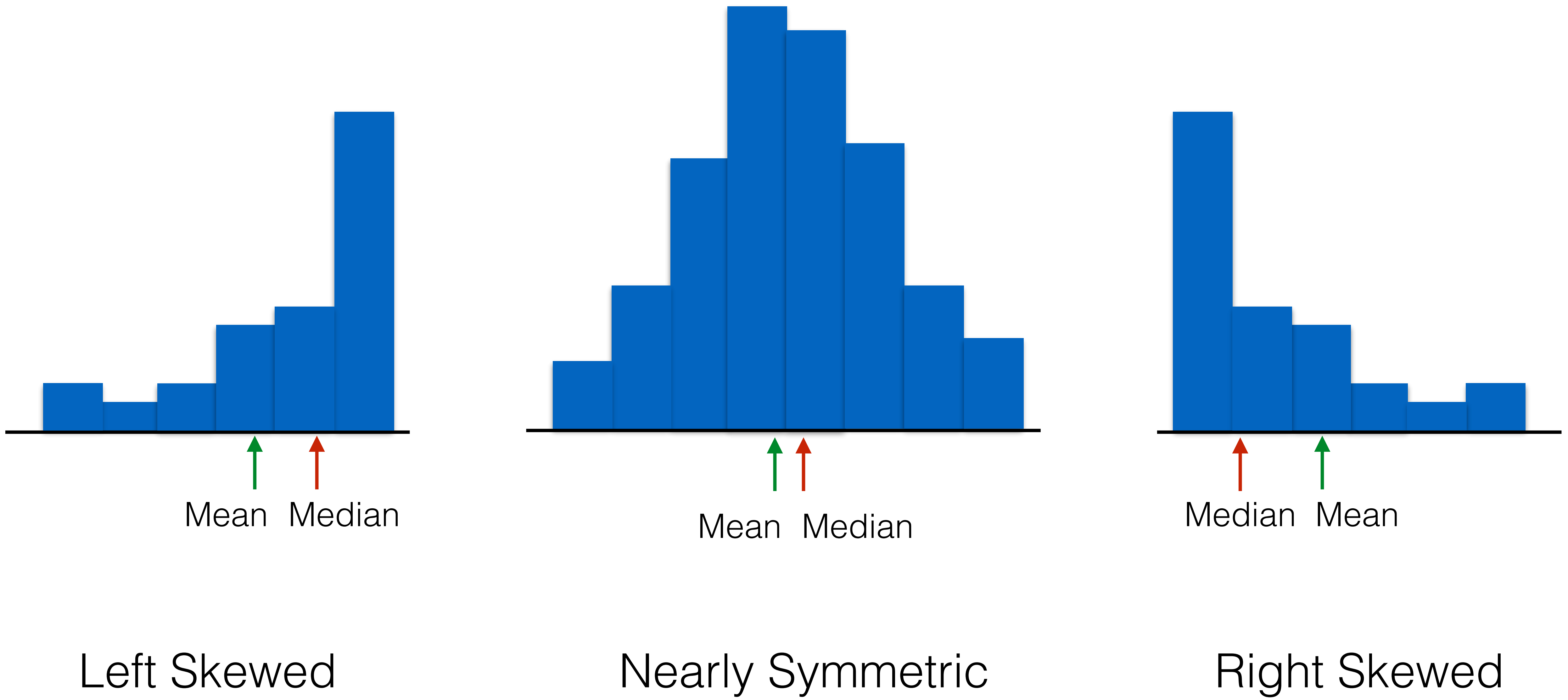
Histograms

- A graphic that gives an idea of the shape of the sample, indicating regions where sample points are concentrated and sparse.



$$\text{Density} = \frac{\text{Relative Freq}}{\text{Class width}}$$

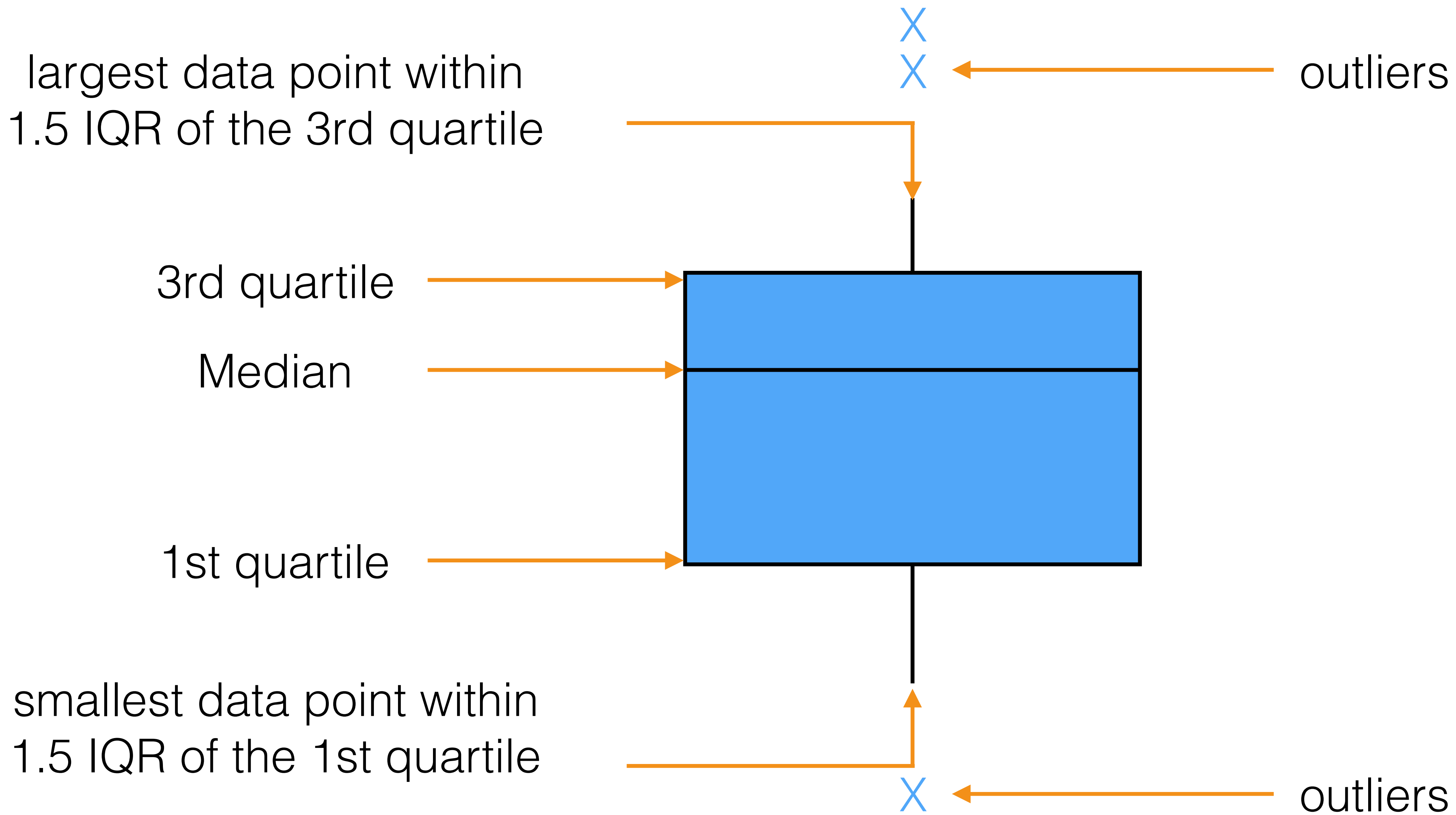
Symmetry and Skewness



Boxplots

- A graphic that presents the median, the 1st and the 3rd quartiles

- Interquartile range (IQR) — the difference between the 3rd quartile and the 1st quartile



Constructing a Boxplot

- Compute the median and the 1st and the 3rd quartiles of the sample. Indicate these with horizontal lines. Draw vertical line to complete the box.

- Find the largest sample value that is no more than 1.5 IQR above the 3rd quartile, and the smallest sample value that is no more than 1.5 IQR below the 1st quartile. Extend vertical lines from the quartile lines to these points.

- Points more than 1.5 IQR above the 3rd quartile, or more than 1.5 IQR below the 1st quartile, are designated as outliers. Plot each outlier individually.

Comparative Boxplots